LAB 2: Full Constraint Closure and Dirac Brackets (Beyond 1+1D)

Primary Constraint (phi1):

```
phi1 = -u0^2 + u1^2 + u2^2 + u3^2 + 1 ≈ 0
```

Secondary Constraint (phi2):

```
phi2 = -2 u0 du0/dt + 2 u1 du1/dt + 2 u2 du2/dt + 2 u3 du3/dt
+ beta * [dtheta/dt * (u0 du0/dt + u1 du1/dt + u2 du2/dt + u3 du3/dt)
         + dtheta/dx * u0 du0/dx + dtheta/dx * u1 du1/dx
         + dtheta/dy * u2 du2/dy + dtheta/dz * u3 du3/dz]
```

Tertiary Constraint Candidate (phi3):

phi3 = d/dt(phi2), contains higher derivatives (ddot(u), dot(theta), etc.)
No new algebraic constraint — phi3 is a dynamical consistency identity.

Constraint Classification:

phi1: second-class
phi2: second-class
phi3: not independent, derived from EOM

Dirac Bracket Summary:

Dirac bracket defined by C_ij = {phi_i, phi_j}, with:
C_12 != 0, C_11 = C_22 = 0

Quantization can proceed using Dirac brackets.
Canonical commutation relations must be replaced accordingly.